```
import pandas as pd
df = pd.read_excel("Cleaned_Retail_Sales_Dataset.xlsx")
df.head()
```

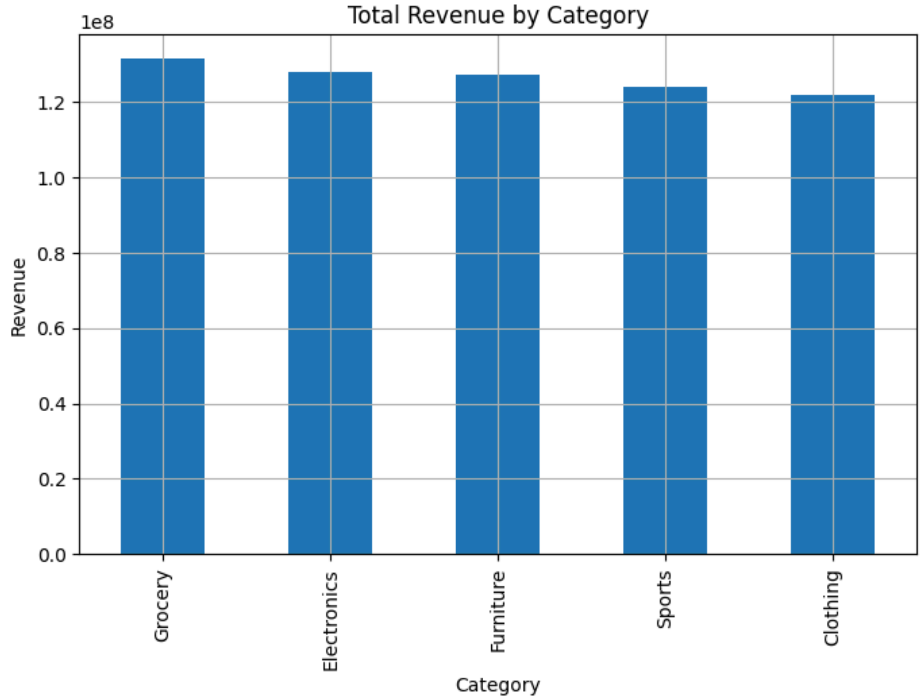
	Order_ID	Order_Date	Category	Product	Quantity	Unit_Price	Revenue	Profit
0	1	2021-01-01	Grocery	Shoes	7	41921	293447	75699.49
1	2	2021-01-02	Sports	TV	7	46261	323827	23925.60
2	3	2021-01-03	Furniture	Table	1	42050	42050	3272.13
3	4	2021-01-04	Sports	Shoes	1	35323	35323	9725.51
4	5	2021-01-05	Sports	Mobile	4	34923	139692	25871.39

Next steps: Generate code with df New interactive sheet

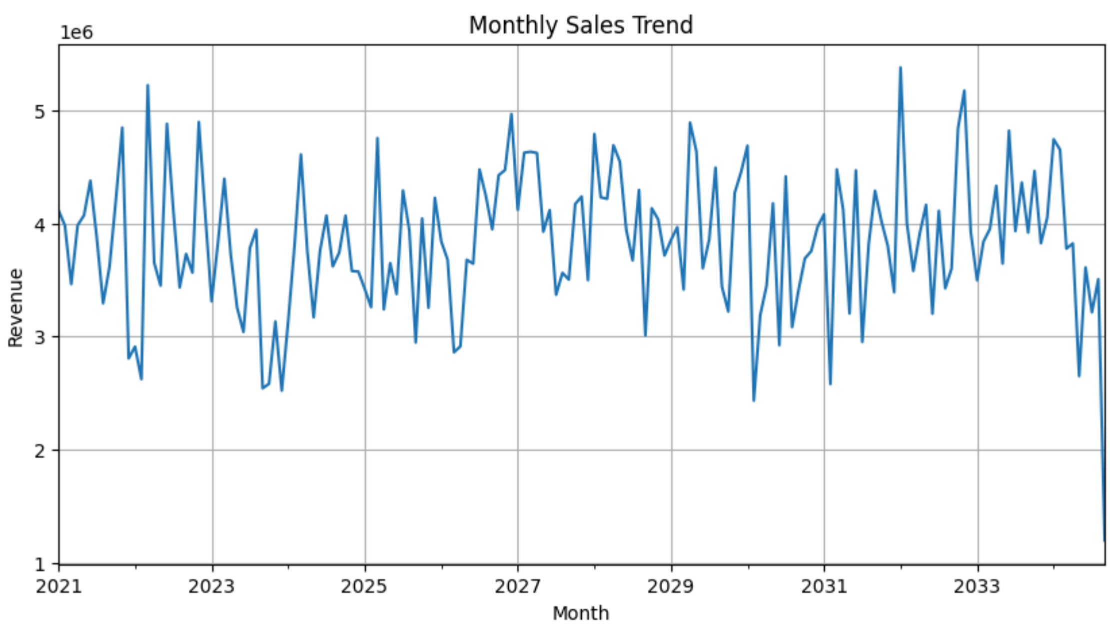
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 5000 entries, 0 to 4999
Data columns (total 8 columns):
 #   Column      Non-Null Count  Dtype
---  ------      --------------  -----
 0   Order_ID    5000 non-null   int64
 1   Order_Date  5000 non-null   datetime64[ns]
 2   Category    5000 non-null   object
 3   Product     5000 non-null   object
 4   Quantity    5000 non-null   int64
 5   Unit_Price  5000 non-null   int64
 6   Revenue     5000 non-null   int64
 7   Profit      5000 non-null   float64
dtypes: datetime64[ns](1), float64(1), int64(4), object(2)
memory usage: 312.6+ KB
```

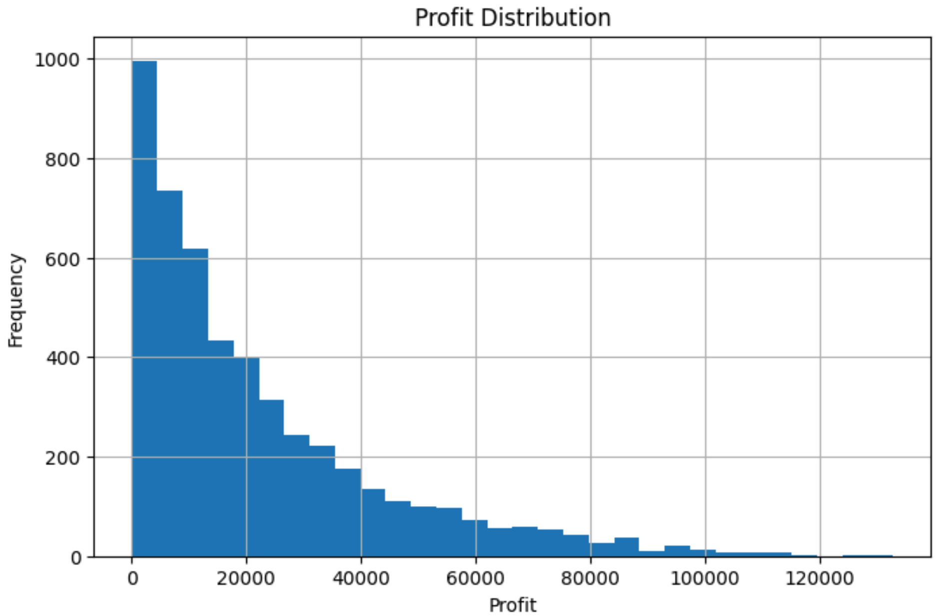
```
import matplotlib.pyplot as plt

top_categories = df.groupby("Category")["Revenue"].sum().sort_values(ascending=False)

plt.figure(figsize=(8,5))
top_categories.plot(kind="bar")
plt.title("Total Revenue by Category")
plt.xlabel("Category")
plt.ylabel("Revenue")
plt.grid(True)
plt.show()
```



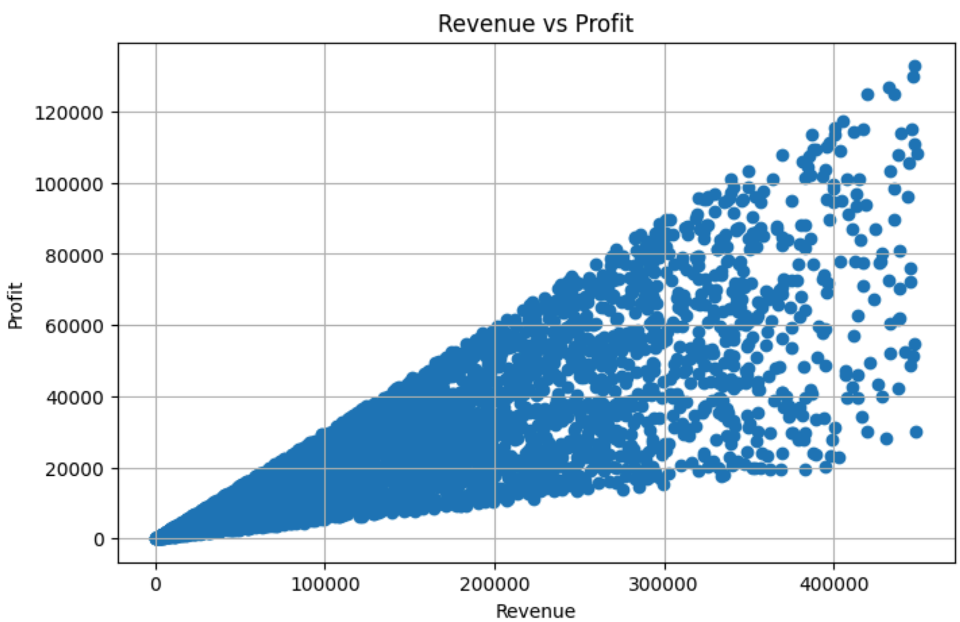

```
df["Order_Date"] = pd.to_datetime(df["Order_Date"])
monthly_sales = df.groupby(df["Order_Date"].dt.to_period("M"))["Revenue"].sum()

plt.figure(figsize=(10,5))
monthly_sales.plot()
plt.title("Monthly Sales Trend")
plt.xlabel("Month")
plt.ylabel("Revenue")
plt.grid(True)
plt.show()
```



```
plt.figure(figsize=(8,5))
plt.hist(df["Profit"], bins=30)
plt.title("Profit Distribution")
plt.xlabel("Profit")
plt.ylabel("Frequency")
plt.grid(True)
plt.show()
```




```
plt.figure(figsize=(8,5))
plt.scatter(df["Revenue"], df["Profit"])
plt.title("Revenue vs Profit")
plt.xlabel("Revenue")
plt.ylabel("Profit")
plt.grid(True)
plt.show()
```




```
max_revenue = df["Revenue"].max()
max_profit = df["Profit"].max()

print("Highest Revenue:", max_revenue)
print("Highest Profit:", max_profit)
```

```
Highest Revenue: 449019
Highest Profit: 132818.8
```

```
df.head()
```